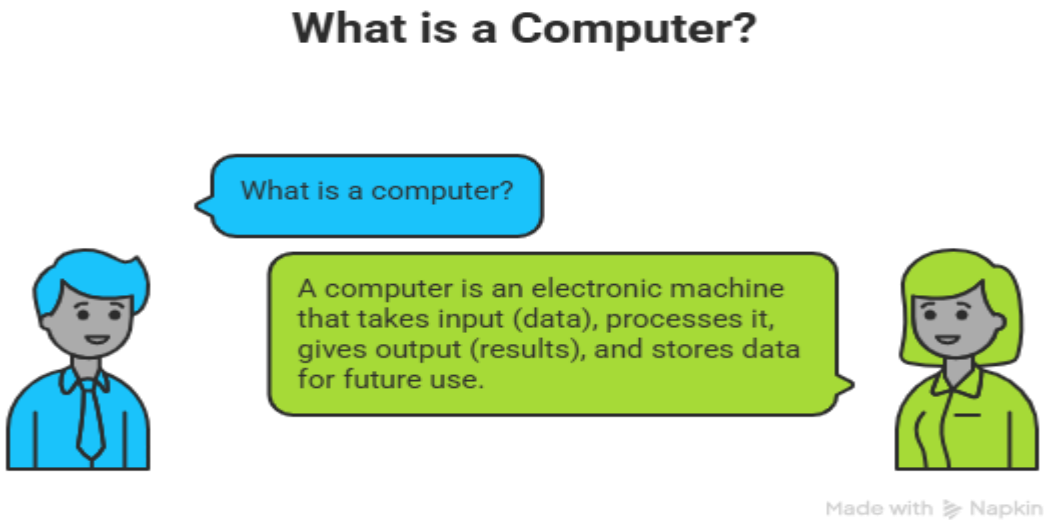
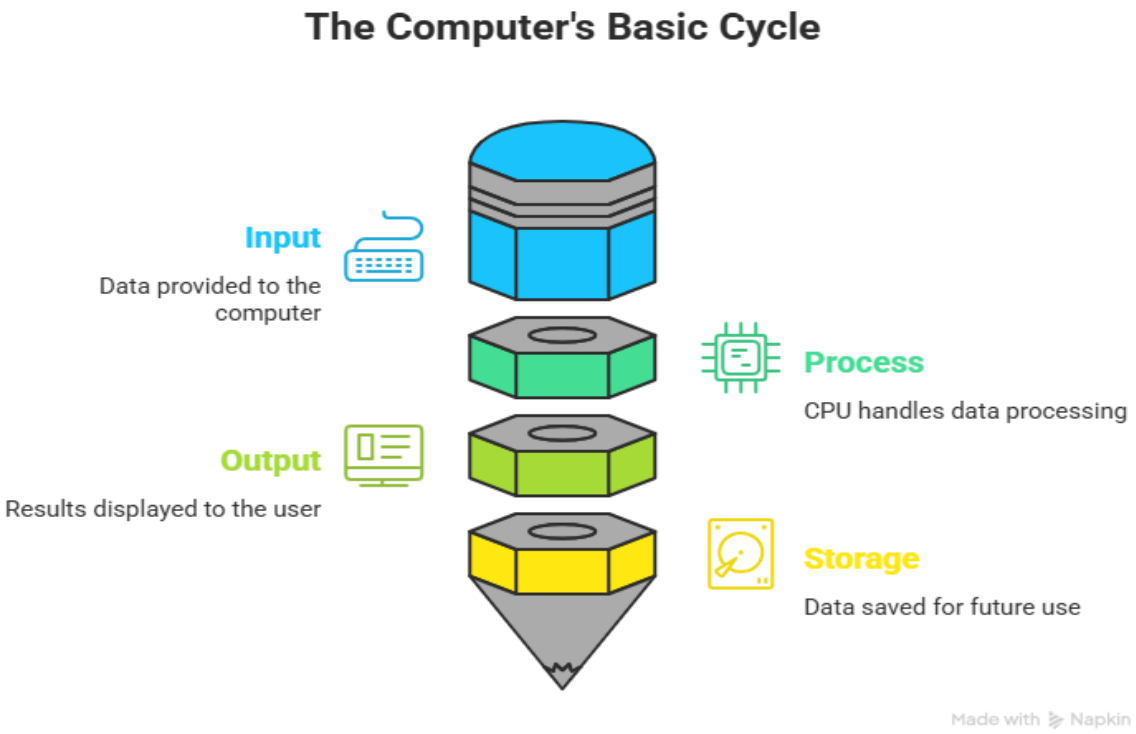
Computer Fundamentals For Beginners

1.What is Computer?

A computer is an electronic machine that takes input(data), processes it gives output(results), and it stores data for future use.



2.How a Computer Works(Basic Cycle)



Every computer works in 4 Steps:

1.Input: Data given to computer.

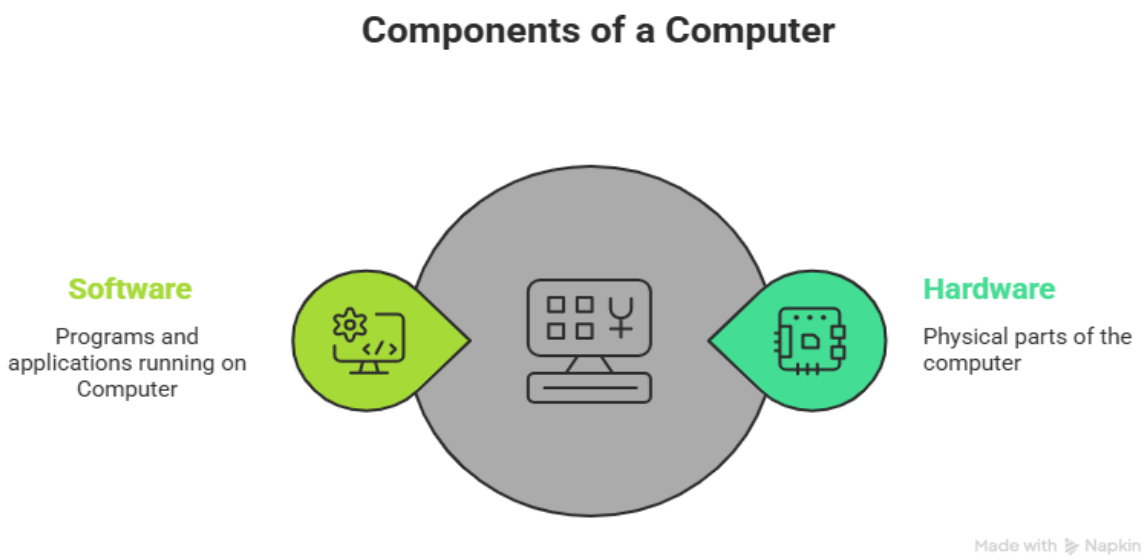
2.Process: Computer processes data using CPU(Central Processing Unit).

3.Output: Result shown to user.

4.Storage: it is used to save the data for future use.

3.Main Components of a Computer

The main components of a computer is Hardware and Software components.



***Hardware(Physical Parts):Things you can touch like Monitor, Keyboard, Mouse and CPU is called Hardware components.**

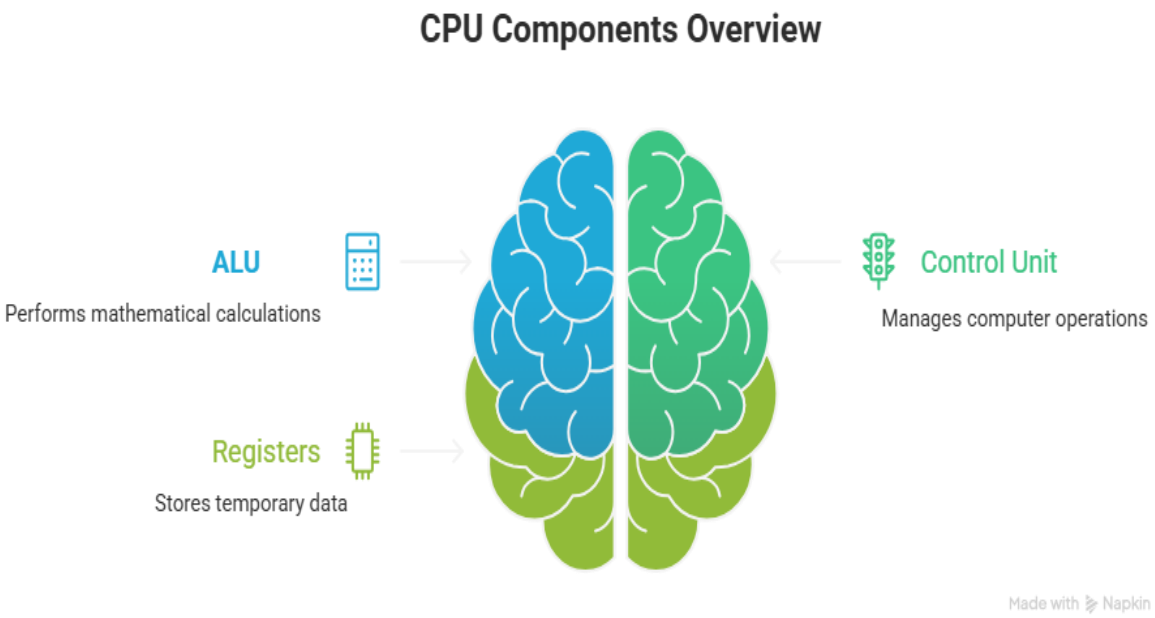
***Software(Programs):Programs that run on computer is called Software components.**

***Without software, hardware cannot work.**

- **Operating System (Windows, Linux, and Macos)**
- **Applications (Chrome, MS Word, and Excel)**

4.CPU(Central Processing Unit)

CPU is the brain of the computer and it performs all calculations and decisions. It has 3 parts:



***ALU (Arithmetic Logic Unit):Performs math calculations.**

***Control Unit(CU):it is used to control the all the operations inside the computer.**

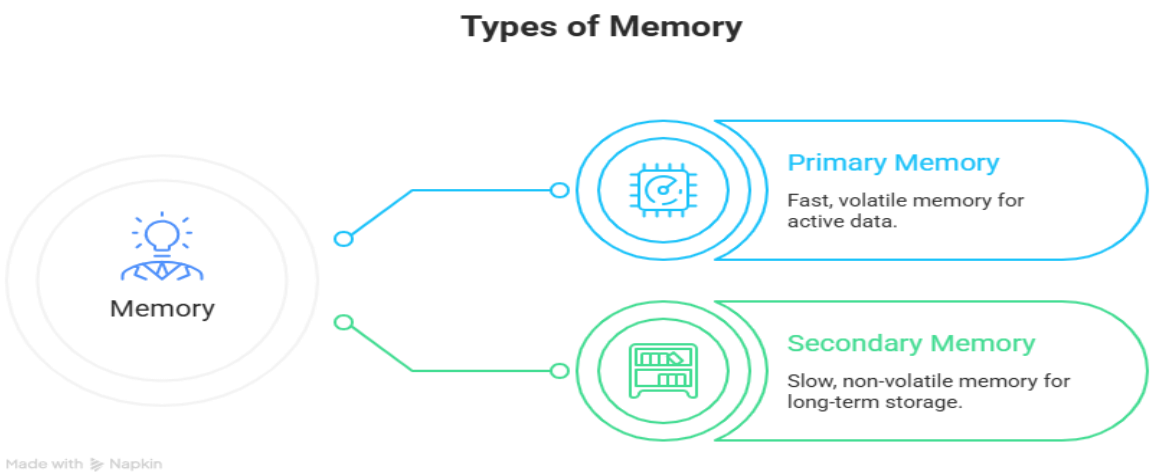
***Registers: register is small temporary memory.**

5.Memory

Memory is the storage area where data and instructions are stored.

Memory are 2 types:

- **Primary Memory**
- **Secondary Memory**

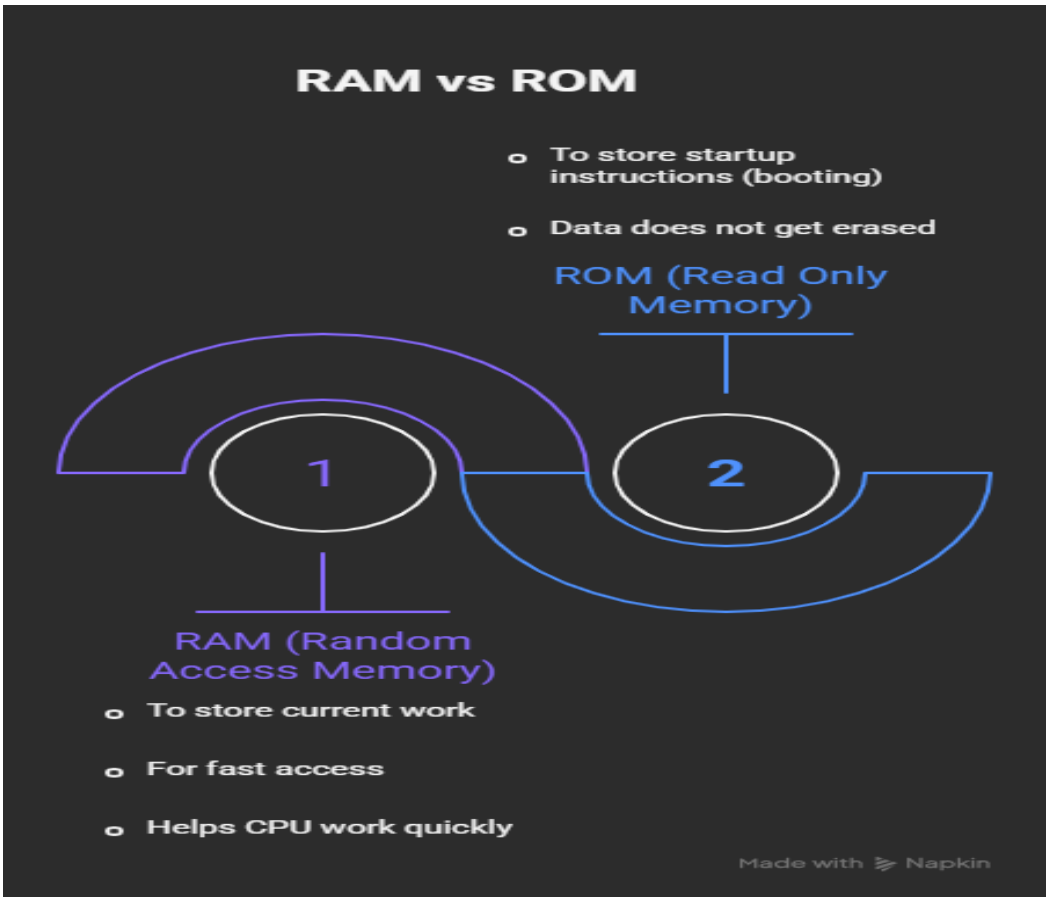


1.Primary Memory(Main Memory):Primary memory is the main memory of a computer that stores data and instructions temporarily while the computer is running. It is directly accessed by the CPU and very fast

Why we use it:

We use primary memory to store the data and programs that are currently being used so the computer can process them quickly.

- ✓ **RAM(Random Access Memory):Temporary memory used while computer is running.[Data lost when power OFF]**
  Why we use it:
  - **To store current work**
  - **For fast access**
  - **Helps CPU work quickly**
- ✓ **ROM(Read Only Memory):permanent memory that stores important instructions**
  Why we use it:
  - **To store startup instructions(booting)**
  - **Data does not get erased.**

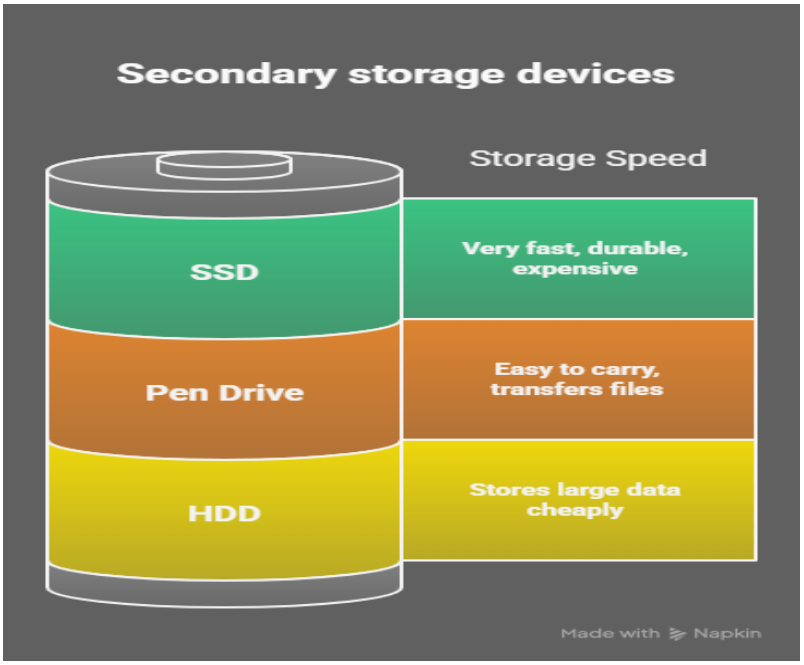


2.Secondary Memory: Permanent storage used to store data for long period of time.

Why we use it:

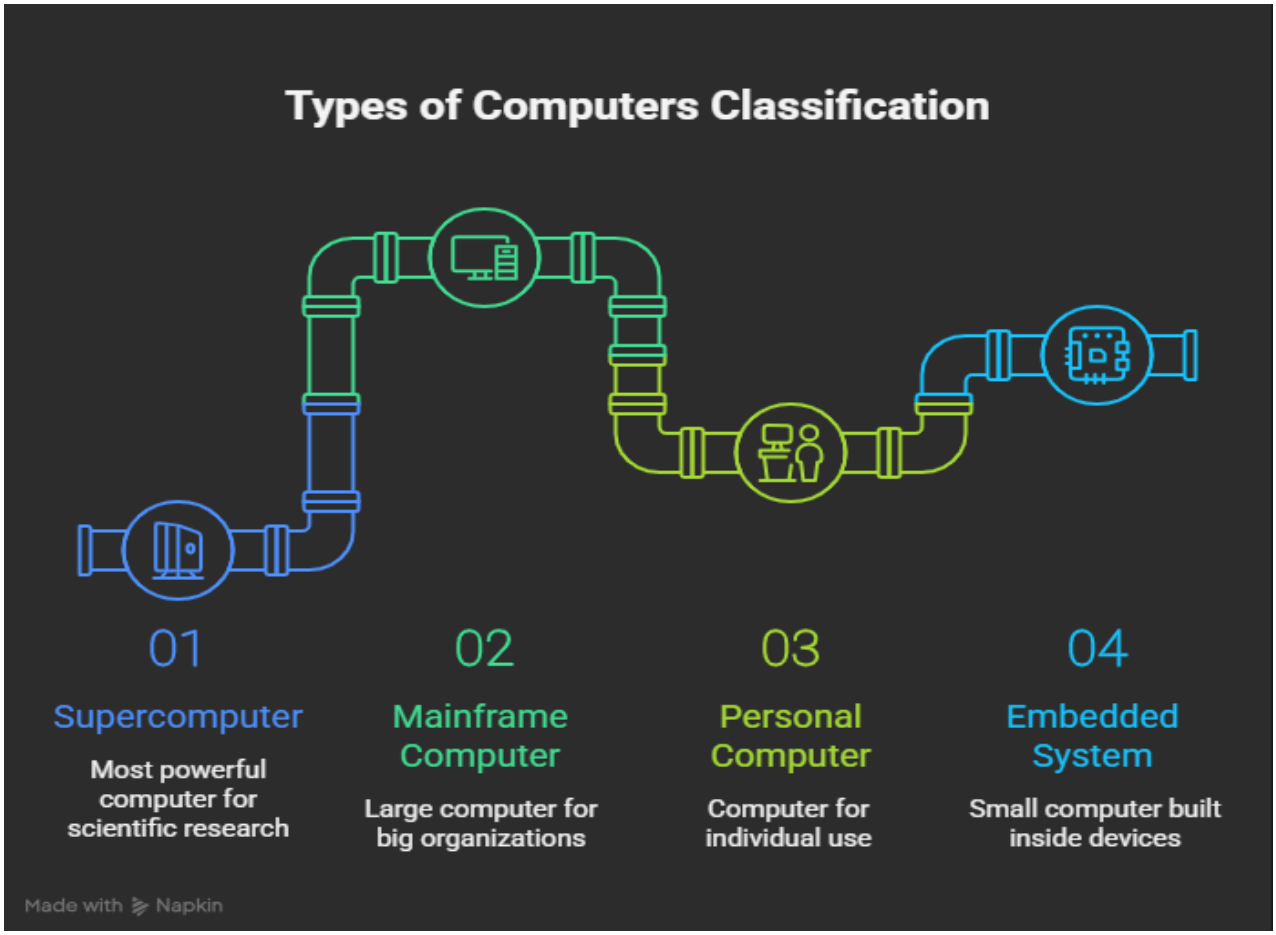
We use secondary memory to store files, software, and data permanently, even when the computer is turned off.

- ✓ **HDD(Hard Disk):it is a secondary storage device that stores data permanently using rotating disk(magnetic storage).**
  Why we use it:
  - **To store large amounts of data(files, videos, software)**
  - **It is cheaper compared to SSD**
  - **Used in computers for long-tearm storage**
- ✓ **SSD(Solid State Drive): an SSD is a storage device that stores data using flash memory(no moving parts).**
  Why we use it:
  - **Very fast performance**
  - **Faster booting and loading**
  - **More durable.**
- ✓ **Pen Drive(USB Flash Drive):it is small portable storage device that uses flash memory and connects through a USB port**
  Why we use it:
  - **Easy to carry data anywhere.**
  - **Used for files transfer between devices.**
  - **Plug and play(no installation needed)**



6.Types of Computers

computers are classified based on size, power, and usage.



Types

1. Supercomputer: it is a most powerful type of computer that can process huge amounts of data very quickly.
   Why we use it:
   It is used for scientific research, weather forecasting, and space studies.
2. Mainframe Computer: it is a large and powerful computer used by big organizations to handle large data.
   Why we use it:
   It is used in banks, airlines, and companies for processing large transactions.
3. Personal Computer(PC):it is a computer designed for individual use, such as desktops and laptops.
   Why we use it:
   It is used for study, office work, internet browsing, and entertainment.

4. **Embedded System: it is small computer built inside a device to perform a specific function.**
   Why we use it
   It is used in machines like washing machines,cars and microwaves to control operations.

7.Basic Computer Terms

- **Bit: it is the smallest unit of data in a computer, and it can be either 0 or 1.**
- **Byte: a byte is a group of 8 bits used to represent a single character like a letter or number.(8 bits=1 byte)**
- **File: a file is a collection of related data stored on a computer.**
- **Program: a program is a set of instructions given to a computer to perform a specific tasks.**